AMERICAN PSYCHOLOGICAL ASSOCIATION

Psychological Review

© 2024 The Author(s)
ISSN: 0033-295X

2025, Vol. 132, No. 3, 744–757
https://doi.org/10.1037/rev0000468

THEORETICAL NOTE

The Double Empathy Problem: A Derivation Chain Analysis and Cautionary Note

Lucy A. Livingston[1, 2], Luca D. Hargitai[3], and Punit Shah[3]
[1] Department of Psychology, Institute of Psychiatry, Psychology and Neuroscience, King's College London
[2] Neuroscience and Mental Health Innovation Institute, Cardiff University
[3] Department of Psychology, University of Bath

Work on the "double empathy problem" (DEP) is rapidly growing in academic and applied settings (e.g., clinical practice). It is most popular in research on conditions, like autism, which are characterized by social cognitive difficulties. Drawing from this literature, we propose that, while research on the DEP has the potential to improve understanding of both typical and atypical social processes, it represents a striking example of a weak derivation chain in psychological science. The DEP is poorly conceptualized, and we find that it is being conflated with many other constructs (i.e., reflecting the "jingle–jangle" fallacy). We provide examples to show how this underlies serious problems with translating theoretical claims into empirical predictions and evidence. To start tackling these problems, we propose that DEP research needs reconsideration, particularly through a better synthesis with the cognitive neuroscience literature on social interaction. Overall, we argue for a strengthening of the derivation chain pertaining to the DEP, toward more robust research on (a)typical social cognition. Until then, we caution against the translation of DEP research into applied settings.

Keywords: derivation chain, empathy, social cognition, social interaction, theory of mind

Supplemental materials: https://doi.org/10.1037/rev0000468.supp

Social interaction is inherently a two-way process, whereby an individual's social cognitive abilities are affected by those of their interaction partner. This applies to a range of "higher order" components of social cognition, including theory of mind and empathy. For example, the ability to understand others' mental states inevitably depends on how "easy" it is to infer your interaction partner's mental state from their behavior (e.g., J. P. Mitchell, 2009; Wu & Mitchell, 2019). However, despite the multidirectional nature of social interaction, most research—both theoretical and methodological—has focused on social cognition in the absence of dynamic social interaction. To prioritize experimental control, most studies measure participants' response to simple social stimuli (e.g., faces, anthropomorphized objects), without capturing social interactions between multiple agents. In recent years, however, there has been a promising push toward understanding the mechanisms underlying dynamic social interaction (see Cañigueral et al., 2022; Hadley et al., 2022; A. F. D. C. Hamilton & Holler, 2023; Redcay & Schilbach, 2019; Sebanz & Knoblich, 2021). There is now a strong awareness that understanding social difficulties in clinical conditions, like autism spectrum disorder (hereafter, autism), requires greater consideration of the two-way nature of social interaction (Bolis et al., 2023; Dingemanse et al., 2023; Milton et al., 2022).

This article was published Online First June 3, 2024.
Lucy A. Livingston https://orcid.org/0000-0002-8597-6525
Luca D. Hargitai https://orcid.org/0000-0003-4536-8848
Punit Shah https://orcid.org/0000-0001-5497-4765
Lucy A. Livingston was supported by the Waterloo Foundation, Luca D. Hargitai is funded by a doctoral studentship from the Economic and Social Research Council, and Punit Shah is supported by the Baily Thomas Charitable Fund. The authors thank Lucy Waldren, Ailidh Finlayson, and Jordan Hembrow for assistance with the analysis code. Lucy A. Livingston and Punit Shah are grateful to Cassius Louis Livingston–Shah for their contribution to the first version of this article.

The research was not preregistered, and there are no data accompanying this Theoretical Note. Analysis code for an automated part of the literature search is provided in the online Supplemental Material.

Lucy A. Livingston played an equal role in conceptualization, investigation, supervision, writing–original draft, and writing–review and editing. Luca D. Hargitai played a lead role in formal analysis and visualization, a supporting role in writing–review and editing, and an equal role in investigation. Punit Shah played an equal role in conceptualization, investigation, supervision, writing–original draft, and writing–review and editing.

Open Access funding provided by University of Bath: This work is licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0; https://creativecommons.org/licenses/by/4.0). This license permits copying and redistributing the work in any medium or format, as well as adapting the material for any purpose, even commercially.

Correspondence concerning this article should be addressed to Lucy A. Livingston, Department of Psychology, Institute of Psychiatry, Psychology and Neuroscience, King's College London, Guy's Campus, London SE1 1UL, United Kingdom, or Punit Shah, Department of Psychology, University of Bath, Claverton Down, Bath BA2 7AY, United Kingdom. Email: lucy.livingston@kcl.ac.uk or p.shah@bath.ac.uk

Autism is a lifelong neurodevelopmental condition, diagnosed behaviorally by the presence of social communication difficulties and repetitive and restricted behaviors and interests (American Psychiatric Association, 2013). Decades of autism research have centered around cognitive theories to explain core autistic behaviors (see Frith, 2012). Theories focusing on social cognition "difficulties" have been especially popular, such as the theory of mind explanations (e.g., Baron-Cohen et al., 2000; Happé, 2015), which postulate that autistic social difficulties arise from a difficulty in understanding the mental states of others. Such social cognitive theories are among the more robust cognitive explanations of autism (see Livingston & Happé, 2021), especially given that comparisons between autistic and neurotypical[1] adults show the largest group differences for social versus other aspects of cognition (see Velikonja et al., 2019, for meta-analysis). This body of research, accordingly, has informed the wider understanding of both typical and atypical social cognitive functions—and their neural correlates (see Happé & Frith, 2014; Kanske & Murray, 2021).

Recent research, however, suggests that social cognitive theories can be limited in predicting autistic and neurotypical people's "real-world" social behaviors (e.g., Bottema-Beutel et al., 2019; Sasson et al., 2020). One of the most popular challenges has been shaped around the "double empathy problem" (DEP). The DEP, coined by Milton (2012), theorizes that autistic people's social difficulties are due to a "mismatch" between autistic and neurotypical people. The DEP theory claims that autistic people do not necessarily have social cognitive difficulties per se but instead struggle to interact with neurotypical people, just as neurotypical people have trouble interacting with autistic people (Davis & Crompton, 2021; Milton et al., 2022; P. Mitchell et al., 2021). It argues, therefore, that the breakdown in social interaction between autistic and neurotypical people is incorrectly attributed to social communication difficulties that lie within the autistic person alone, rather than two-way social cognitive difficulties. Indeed, the strongest proponents of the DEP theory imply that social cognitive deficits cannot be said to exist in autism (e.g., Chapman, 2019; Chown et al., 2020). These are major claims that warrant investigation. If correct, they have obvious potential to revolutionize traditional understandings of autism and other neurodevelopmental, neuropsychological, and psychiatric conditions characterized by social difficulties.

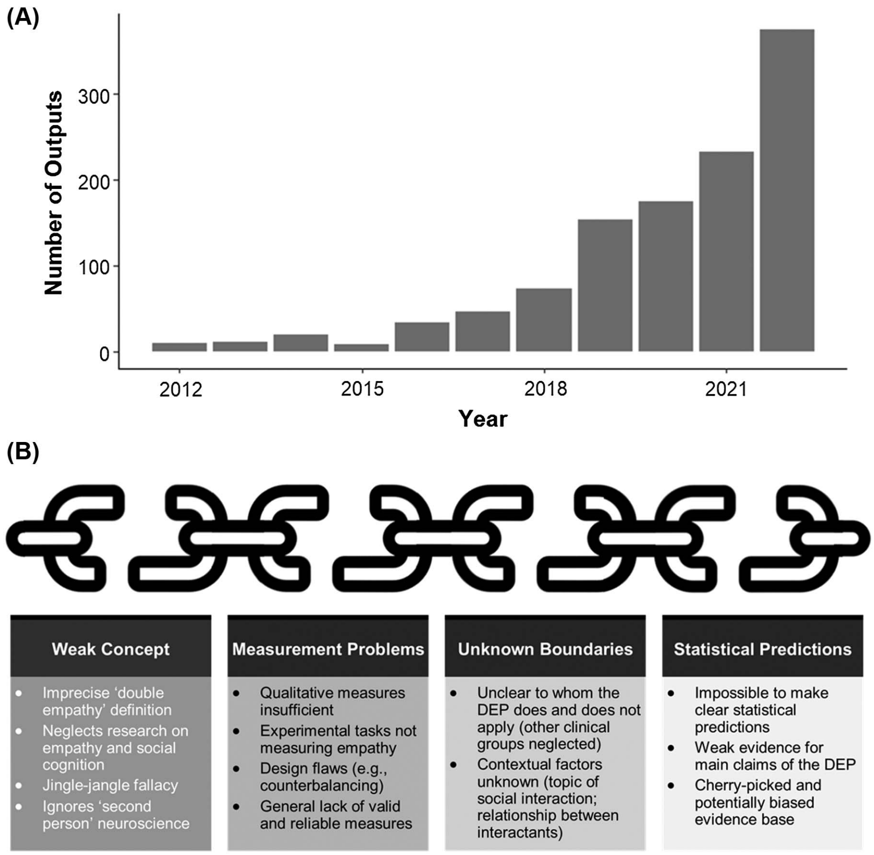
The double empathy concept is rapidly growing in popularity (Figure 1A), with a sharp increase in studies claiming to support the phenomenon (e.g., Chen et al., 2022a; Crompton, Hallett, et al., 2020; Hand et al., 2023; Morrison, DeBrabander, Jones, Faso, et al., 2020). Researchers are also invoking the DEP to explain complex results on social cognition that are difficult to interpret, or when discussing null findings in studies comparing autistic and neurotypical people (see Gernsbacher & Yergeau, 2019). Further, its reach now extends well beyond academia, with the concept permeating everyday parlance in nonacademic settings (e.g., by the U.K.'s leading autism charity, the National Autistic Society; see Milton, 2018). Most strikingly of all, putative evidence for the DEP has been suggested to "challenge the diagnostic criterion that autistic people lack the skills to interact successfully" (Crompton, Ropar, et al., 2020, p. 1704), alongside other calls for a shift in clinical practice based on the DEP (e.g., Milton et al., 2022; Nicolaidis et al., 2019). Training for autism professionals is integrating DEP theory (see Milton, 2018; Milton et al., 2018), resources are being created to teach children and young people about it (Crompton et al., 2021), and clinical and educational interventions are being suggested to overcome the DEP between autistic and neurotypical people (e.g., Camilleri et al., 2023; Chapman & Botha, 2023; Chapple et al., 2021; L. G. Hamilton & Petty, 2023; Jellett & Flower, 2023).

We do not doubt that there are fundamental truisms underlying the DEP phenomenon, particularly in light of several decades of research in social psychology on how social contexts and group identities fundamentally shape human interaction (see Freese & Burke, 1994; House, 1977). Additionally and importantly, we do not wish to dismiss the progress that has been made to date. Yet, in the face of such widespread academic and public interest in the DEP, it is surprising that the DEP concept has never been formalized as a theory as per usual standards in the psychological and cognitive sciences (see Eronen & Bringmann, 2021). There is no detailed formulation, with central assumptions and concrete predictions for empirical testing, yet the DEP is regarded as a robust theory by many scientists, scholars from other disciplines (e.g., education), clinicians, and other practitioners. The DEP theory must now be subjected to greater scrutiny, necessitating a close examination of its main claims and the measures being used. This will uncover the evidence base, if any, for the DEP, helping to establish whether ongoing changes in research practice and clinical, educational, and social policies are warranted. In this article, we will critically evaluate the DEP concept, drawing on autism research, that is, where the bulk of research on the DEP has been concentrated. Specifically, we propose to assess the strength of the various elements of the DEP "derivation chain" (Meehl, 1990b), that is, the links connecting the DEP concept with the currently available methodological approaches and empirical data. This is not an exhaustive or systematic review of the DEP literature, and the studies referenced are illustrative examples. Indeed, given the loose and varied manner that the concept of "double empathy" has been defined and operationalized in the literature (see the Fuzzy Theoretical Concept section), it is currently not possible to conduct a systematic review with appropriate precision. We propose that strengthening the derivation chain in this field will ultimately ensure that the real-world impact of the DEP theory is more likely to be realized.

In one of many highly influential articles, Meehl (1990b) proposed that a strong "derivation chain"—linking theoretical claims to empirical predictions from data—is required to advance psychological knowledge (see also Scheel, Tiokhin, et al., 2021). If one imagines that the former sits at the beginning of the chain and the latter at the end of the chain, there are various intermediary elements that are necessary to pass through. These elements include but are not restricted to (a) valid and reliable measurement tools, for (b) adequate testing of the theory's "boundaries" (i.e., in which contexts the theory does [not] apply), and (c) robust statistical predictions (i.e., use of statistical tests to falsify theoretical predictions). If any of these elements are not robust, the derivation chain is weakened, and the confidence with which knowledge can be advanced from any arising data is diminished. Following this Meehlian approach, we evaluate the DEP theory by assessing the strengths and weaknesses of the elements in the derivation chain (Figure 1B), starting with the theoretical concept.

[1] Neurotypical is a widely used term for individuals who would be considered "typically developing," that is, do not have a neurodevelopmental condition, such as autism (Fletcher-Watson et al., 2021).

Figure 1
Research on the Double Empathy Problem



Note. (A) Scholarly outputs including "double empathy." Google Scholar analysis on August 21, 2023, for outputs including "double empathy" as a search term. This shows an average year-on-year increase of over 60% between 2012 (i.e., when the term was coined) and 2022. The code for the Google Scholar search is available in the online Supplemental Material. The code for the Google Scholar search is based on Pold87/Academic-Keyword-Occurrence: First Release (Version v1.0.0) [Computer software], by V. Strobel, 2018, Zenodo (https://doi.org/10.5281/zenodo.1218409). Copyright 2018 by Volker Strobel. (B) The double empathy problem (DEP) derivation chain. A depiction of the weak derivation chain for the DEP, including broken links between each component.

Fuzzy Theoretical Concept

A successful derivation chain first requires sound theoretical concepts (Scheel, Tiokhin, et al., 2021). From its inception, the concept of the DEP, stemming from sociology, does not readily map onto the fields of psychology and cognitive science that are now developing and studying it apace. Milton (2012) initially conceived the DEP as "a disjuncture in reciprocity between two differently disposed social actors which becomes more marked the wider the disjuncture in dispositional perceptions of the lifeworld" (p. 884). In a recent update, Milton et al. (2022) referred to it as a "breakdown in mutual understanding … and hence a problem for both parties to contend with, yet more likely to occur when people of very differing dispositions attempt to interact" (p. 1901). When contrasted against analogous frameworks and theories in social cognitive science (e.g., Bird & Viding, 2014; Cañigueral et al., 2022), these definitions of the DEP lack precision as the starting point of a derivation chain. Most obviously, despite encapsulating "empathy" as the central focus, the DEP talks loosely about "disjuncture" and "shared understanding," but not any well-recognized definitions of empathy. More widely, it underspecifies which, if any, specific social cognitive processes are involved, even in the most up-to-date literature (Milton et al., 2022, 2023).

This conceptual ambiguity is particularly puzzling given that a substantial body of research has gone to painstaking lengths to differentiate empathy from other closely related social cognitive processes, such as social reciprocity, emotion recognition, and theory of mind (see Happé et al., 2017; Schurz et al., 2021). While serious challenges with conceptualizing social cognition and its subcomponents, such as empathy, are ongoing (e.g., Olderbak & Wilhelm, 2020; Quesque & Rossetti, 2020; Stellar & Duong, 2023), one widely accepted definition of empathy refers to sharing the affective state of another individual (e.g., experiencing pain when you observe someone else in pain). Both theoretical (e.g., Bird & Viding, 2014; Gonzalez-Liencres et al., 2013) and empirical (e.g., Kogler et al., 2020; Yang et al., 2018) work also indicates that empathy is a multidimensional construct (e.g., with cognitive and affective components). This research highlights, for example, that empathy is distinguishable from the initial identification of another person's emotion (i.e., emotion perception/recognition) and understanding of their nonaffective mental states, such as thoughts, intentions, and beliefs (i.e., theory of mind).

Because "double empathy" was never precisely defined, there is now widespread evidence of the "jingle–jangle" fallacy in the literature. The DEP is being used to refer to various constructs (i.e., jingle), as well as different concepts referred to as double empathy (i.e., jangle). This problem is not unique to the DEP and reflects a wider problem in social cognition research (e.g., Clutterbuck et al., 2021, 2022; Murphy et al., 2022; Quesque & Rossetti, 2020), including the empathy literature specifically (see Olderbak & Wilhelm, 2020, for detailed discussion). Across a range of studies, the term "DEP" is used, despite explicitly describing and measuring a range of entirely different social cognitive constructs. This includes but is not limited to research on theory of mind (e.g., Alkire et al., 2023), emotion recognition (e.g., Hand et al., 2023), "shared understanding" (e.g., Chapple et al., 2021; Heasman & Gillespie, 2019; G. L. Williams et al., 2021), perspective taking (e.g., DeNigris et al., 2018; Heasman & Gillespie, 2018), social perception (e.g., Jones et al., 2021; Morrison, DeBrabander, Jones, Faso, et al., 2020), and observable social behavior (e.g., Chen et al., 2021, 2022a). At the same time, a similar range of social cognitive constructs are being described in the very definition of the DEP theory (e.g., Chapman, 2019; Chown, 2014; Milton, 2012; Milton et al., 2022), thus incorrectly implying that all these terms are synonymous with the DEP. Overall, we argue that the DEP phenomenon does not currently account for any existing definitions of empathy or related social cognitive constructs, leading to a fuzzy theoretical concept at the start of its derivation chain.

On a final note regarding conceptual ambiguity, we draw attention to the "dialectical misattunement hypothesis" of clinical conditions, including autism (Bolis et al., 2018, 2021, 2023). Despite arising separately to the concept of the DEP, with its origins in social cognitive neuroscience, it bears striking similarity to the DEP theory in its definition: It "rethinks a psychiatric condition, such as [autism], not merely as a disorder of the individual brain but also as cumulative misattunement between persons" (Bolis et al., 2018, p. 357). More widely, the theoretical conceptions of "second-person neuroscience" and "second-person neuropsychiatry" have been around for a decade (e.g., Schilbach, 2016; Schilbach et al., 2013), including carefully developed ideas that draw on ecological psychology (e.g., Knoblich & Sebanz, 2008). This research provides a detailed, neuroscientifically grounded approach to social interaction. There is a rich breadth and depth to this approach, including perspectives on how to reconcile first- and second-person neuroscience (e.g., Longo & Tsakiris, 2013), developmental perspectives (e.g., Lewis & Stack, 2013), as well as computational approaches (e.g., Lehmann et al., 2023). Almost none of this research has been considered as part of the theoretical development of the DEP. The extent to which these theoretical positions should be considered distinct, one and the same, and/or with overlapping features, requires further clarification. We return to this as a crucial direction for future research.

Overall, we contend that the DEP falls at the first hurdle, at the starting point of the derivation chain. Due to a paucity of theoretical groundwork, which neglects the literature on theories of social cognition and interaction, the DEP lacks coherence with and differentiation from other constructs—that is, it fails to meet key criteria for good conceptualization in the social sciences (as per Gerring, 1999). There does not appear to be a clear consensus about what the DEP is and, crucially, what it is not. This is not only problematic in and of itself but underlies the following cascade of problems with other aspects of the derivation chain.

Measurement Challenges

Without a sound concept and theory as a foundation, there is a domino effect on measurement and the rest of the derivation chain (see Scheel, Tiokhin, et al., 2021, for examples). That is to say, if we do not understand exactly *what* the DEP is, it is difficult to develop methods to ensure its valid and reliable measurement. It could therefore be argued that empirical testing should not proceed. Empirical research on the DEP is, however, being conducted apace. Practically speaking, we do not foresee this stopping or slowing down. We therefore propose that the existing body of empirical research warrants inspection to see if any lessons can be learned for the future. Across this literature, almost entirely on autism, we find four general types of study that have been taken to measure the DEP.

First, researchers have asked small groups of people about their lived experiences relevant to the DEP (e.g., Camus et al., 2022; Chapple et al., 2021; Crompton, Hallett, et al., 2020; DeNigris et al., 2018; Heasman & Gillespie, 2018). For example, aiming to tap into the DEP, Chapple et al. (2021) analyzed interviews with four autistic–neurotypical dyads, who had read and discussed a novel together over a 1-week period. These studies generally find that autistic people find interactions easier and more relatable with other autistic compared to neurotypical people. Such qualitative research can provide crucial insights into novel phenomena and play a role in early theory development and the creation of quantitative measurement tools (e.g., Livingston et al., 2020; Ricci et al., 2019).

Crucially, however, almost none of these studies interviewed participants directly about their empathic or social cognitive experiences more generally. This is compounded by the fact that qualitative measurements and analyses are subjective. They are inherently influenced by researchers' own perspectives and, without asking precise questions, do not produce data from which clear hypotheses can be falsified. Taken together, we argue that qualitative measures and arising lived experience data, when used in isolation and without alignment with existing social cognitive theories, are not sufficient for measuring the DEP. Interestingly, Meehl derived insights from clinical cases and experiential data to inspire theoretical development (Burton & Harris, 1947). Equally, Meehl recognized that this must be coupled with psychometrically robust measurement, especially for clinically relevant constructs (e.g., Meehl, 1954; Meehl & Rosen, 1955). We highlight this point to guard against claims that the Meehlian approach to derivation chains is not consistent with approaches to theory development that include lived experience data. Indeed, the compendium of Meehl's work (e.g., on schizophrenia; Lenzenweger, 2006; Meehl, 1962, 1990a) provides examples of how such approaches, when informing the development of other methods, can advance knowledge.

Second, most studies on the DEP have involved directly studying social interactions between autistic and neurotypical people. These have either observed interaction in real-world settings (e.g., Chen et al., 2021, 2022a, 2022b; Heasman & Gillespie, 2019) or the lab (e.g., Alkire et al., 2023; Crompton, Ropar, et al., 2020; Crompton, Sharp, et al., 2020; Jones et al., 2021; Morrison, DeBrabander, Jones, Ackerman, & Sasson, 2020; Morrison, DeBrabander, Jones, Faso, et al., 2020; G. L. Williams et al., 2021). In particular, comparing same (i.e., autistic–autistic, neurotypical–neurotypical) with mixed (i.e., autistic–neurotypical) neurotype[2] interactions is commonplace. However, in line with the imprecise conceptualization of the DEP, it is unclear which social processes, exactly, were measured in these studies. An illustrative example of a naturalistic observation study is Chen et al. (2022b), which measured basic interaction between six autistic and six neurotypical adolescents at a school club to plot social networks. An illustrative example of a lab-based social observation is Morrison, DeBrabander, Jones, Ackerman, & Sasson (2020): Following a 5-min unstructured conversation between male participants in same or mixed neurotype dyads, participants self-reported their impressions of the interaction (e.g., rapport) and conversation partner (e.g., personality traits). These studies generally measure some form of social communication and interaction, but it seems unclear how they have measured double empathy per se. For example, the studies could have been designed to observe and/or test participants in how much or how accurately they shared their interaction partners' emotional states after a conversation on emotionally sensitive vignettes (i.e., measuring empathy as affect sharing).

The most sophisticated study that attempts to measure the DEP is a preregistered study by Crompton, Ropar, et al. (2020). They investigated how accurately details from a story were transmitted across eight-person "diffusion chains," that is, Person 1 tells the story to Person 2, who tells the story to Person 3, and so forth (i.e., drawing on Henderson, 1903, and Bartlett's, 1932, classical methods). Specifically, they compared diffusion chains of only autistic people, only neurotypical people, or mixed neurotypes (i.e., autistic–neurotypical–autistic–neurotypical, etc.). Here, the "success" of the social interaction was measured more precisely compared to other studies (i.e., accuracy in retention of story information). However, other than measuring general communication skills by proxy, the underlying social cognitive mechanisms were left uninvestigated. For example, to specifically tap into empathy, an experimental manipulation to compare how well emotional versus nonemotional stories are shared could have been used, with the former being the empathy condition and the latter the control condition. Whether extraneous, nonsocial processes (e.g., acoustic features and/or clarity of speech) could account for how well information was transmitted across the diffusion chains was also not considered, despite research showing that autistic people have distinctive prosodic features (see Asghari et al., 2021, for a meta-analysis).

More generally, given the potential importance of individuals' neurotype, it is surprising that each diffusion chain was started by the researcher (of an unknown neurotype), who was seemingly not blind to the aims of the study. Each mixed neurotype chain also always began with a neurotypical participant. It is therefore a possibility that Crompton, Ropar, et al.'s (2020) results were influenced by both the researcher's neurotype and behavior. A more optimal design, in the future, could employ counterbalancing to ensure that mixed neurotype diffusion chains are also initiated by autistic people. To exert further experimental control, the diffusion chain could be initiated by a nonhuman agent (e.g., computerized reading of the story at a precise frequency and volume), or the researcher could be blinded to the aims of the study and their behavior recorded and analyzed (e.g., to ensure the story is read at a consistent speed and volume across conditions). Alternatively, simply having the first participant in the diffusion chain read the story, following the earliest (Henderson, 1903) and more recent (e.g., Breithaupt et al., 2022) studies using such methods, would likely reduce experimenter bias. Overall, this example serves to illustrate how methodological limitations make it difficult to draw firm conclusions from even the most sophisticated studies on the DEP.

Third, a handful of tasks have been regarded as measures of the DEP (though never being explicitly designed for this purpose) by zooming in on specific social cognitive processes (e.g., emotion recognition, theory of mind). These studies investigated how autistic versus neurotypical people respond to social stimuli generated by autistic versus neurotypical actors. For example, in Brewer et al. (2016), 14 autistic and 13 neurotypical adults were tasked with recognizing emotions in facial expressions that were portrayed by autistic and neurotypical people. Similarly, Edey et al. (2016) adapted the Frith–Happé Animations Test to investigate how well 22 autistic and 24 neurotypical adults identified mental states from animations of interacting triangles. Crucially, some animations had been generated by autistic people, whereas others were generated by neurotypical individuals. The measures used in these studies enabled a much more precise investigation into the social cognitive processes potentially underlying the DEP.

However, following the same fundamental limitation with other measurement approaches, empathy itself was not measured in these studies. This is not surprising as the studies were never explicitly designed to measure the DEP. A further limitation is that such tasks are limited in studying the DEP as a process of social interaction. They involve studying social cognition "offline" and do not emulate the dynamic two-way nature of social interaction (see Redcay & Schilbach, 2019). A final issue with these social cognition tasks is that there is considerable debate about their general validity, irrespective of their use in relation to the DEP. For example, many popular theory of mind tasks often have poor control conditions, require subjective scoring (see Livingston et al., 2019), and have poor psychometric properties (e.g., Andersen et al., 2022; Kittel et al., 2022). Put differently, challenges with measuring social cognition (see Olderbak & Wilhelm, 2020) are being inherited in DEP-related research.

A fourth set of studies that have been cited in support of the DEP theory use methods that do not attempt to capture the shared experience between autistic and neurotypical people. Specifically, several studies focus on neurotypical people's perceptions or understanding of autistic people and their behaviors, without explicitly testing the reverse association (e.g., Alkhaldi et al., 2019; Sasson et al., 2017; Sheppard et al., 2016; but see DeBrabander et al., 2019). Other studies have simply compared autistic and neurotypical people—for example, in their social preferences, behaviors, or cognition—outside of two-way interaction yet have been referenced as evidence for the DEP (Finke, 2023; Grossman et al., 2019; Hand et al., 2023). For example, Finke (2023) investigated autistic and neurotypical adults' friendships, finding several group differences (e.g., autistic people are less likely to have in-person vs. online friends). While such study designs shed some light on how differing neurotypes approach social interactions, we argue that these approaches cannot be taken as measures of the two-way DEP between individuals.

[2] Neurotype is a category distinction, used widely in the DEP literature, to separate neurotypical people from individuals with a neurodevelopmental condition, such as autism (Fletcher-Watson et al., 2021).

Overall, a diverse range of tasks have aimed to measure the DEP and/or been interpreted as measures of the DEP. Many of these approaches have been taken from other fields (e.g., social and cognitive psychology), making assumptions about their validity and suitability for (a) studies in the autistic population and (b) studies within dynamic social interaction between autistic and neurotypical people. None have directly measured the "empathy" part of double empathy insofar as how empathy is currently measured in social cognitive science (e.g., Mackes et al., 2018; Singer et al., 2004; Zaki et al., 2009). This follows a general trend in autism research, and clinical science more widely, whereby measures developed in the general population are used without fully considering their conceptual and psychometric validity in clinical populations and for testing clinically relevant theories (e.g., see example in Z. J. Williams & Gotham, 2021). Even where measures used in studies regarding the DEP were specifically developed for this purpose (e.g., Alkire et al., 2023; Chen et al., 2021, 2022a, 2022b; Sasson et al., 2017), there has been little effort to formally quantify their validity (e.g., convergent/divergent validity) and other psychometric properties (e.g., measurement invariance, test–retest reliability). Several empirical studies related to the DEP have also relied heavily on data collected in student populations (e.g., DeNigris et al., 2018; Jones et al., 2021; Morrison, DeBrabander, Jones, Ackerman, & Sasson, 2020) and/or largely male samples (e.g., Chen et al., 2022b; Finke, 2023; Heasman & Gillespie, 2018; Jones et al., 2021; Morrison, DeBrabander, Jones, Ackerman, & Sasson, 2020; Morrison, DeBrabander, Jones, Faso, et al., 2020). To have confidence that DEP measures are truly measuring DEP, validity should be demonstrated across multiple settings and populations. Overall, we argue that several, rudimentary procedures for developing and administering measurement tools are lacking in DEP research. This is likely to be a cause and consequence of poor theorizing.

Unknown Theory Boundaries

A critical but overlooked part of theory development is establishing the theory's boundaries, that is, the precise parameters in which the theory applies. Without this, it is difficult to formulate specific hypotheses and, consequently, interpret unclear and/or null results. Given the aforementioned concerns with the basic conceptualization and measurement of the DEP, it could be argued that continuing our evaluation at this stage of the derivation chain is futile. Nevertheless, we briefly highlight two key boundary issues that seem particularly obvious in relation to the DEP, with a view to start thinking about future directions.

First, it will be important to establish exactly who the DEP theory does and does not apply to. Most empirical research has focused on the DEP between autistic and neurotypical people, while overlooking many other conditions where social abilities may be atypical, such as schizophrenia (Green et al., 2015), attention-deficit/hyperactivity disorder (Bora & Pantelis, 2016), neurodegenerative disorders (Christidi et al., 2018), social anxiety (Pearcey et al., 2021), and depression (Weightman et al., 2014). The selective focus on autism is surprising given that Milton (2012) originally put forward the DEP as a general mismatch in "disposition" between individuals. Establishing whether the theory can apply to any two individuals, irrespective of diagnosable conditions or other ingroup–outgroup memberships, is essential. It will also be necessary to clarify what, exactly, is understood as "disposition." Does disposition comprise social cognitive processes, like empathy, specifically? Or does it also encapsulate more general characteristics that could influence the interaction, including the interaction partners' social judgments of one another (e.g., personality traits, sociodemographic factors, ethnicity)? This raises a further question as to whether the DEP lies on a continuum—whereby any two individuals can experience even a low level of mismatch—or whether there in fact needs to be a critical threshold of a mismatch for the DEP to be assumed. In other words, testing the theory's boundaries will reveal if there is something qualitatively distinct about the DEP compared to everyday misunderstandings in human social interaction. Constructing and addressing such questions will be crucial to enable reliable predictions about when we would and would not expect to observe the DEP.

Second, the role of context in the DEP remains unclear. For example, in relation to autism, the extent to which knowing another person's diagnostic status influences the two-way interaction has received no empirical attention in a systematic way. Some studies disclose this information to participants (e.g., Crompton, Ropar, et al., 2020; Crompton, Sharp, et al., 2020; Heasman & Gillespie, 2018), while others do not (e.g., Brewer et al., 2016; Sheppard et al., 2016). There are also certain studies where there is a mix of participants who are and are not aware of their interaction partner's diagnostic status (e.g., Jones et al., 2021; Morrison, DeBrabander, Jones, Ackerman, & Sasson 2020). It is unfortunate that this has not been systematically investigated thus far given that diagnostic information might be crucial in influencing social interaction, potentially in unforeseen ways. For example, there is evidence to suggest that diagnostic disclosure is associated with how individuals with various clinical conditions or "hidden disabilities" are perceived by others (e.g., DeBrabander et al., 2019; O'Connor et al., 2022; Sasson & Morrison, 2019).

Another aspect of context that has been largely unexplored is the conversational topic and nature of the social interaction. To our knowledge, most studies on the DEP using dyads have involved superficial "social" situations, such as brief unstructured conversations between participants, often meeting for the first time (e.g., Morrison, DeBrabander, Jones, Ackerman, & Sasson, 2020; Morrison, DeBrabander, Jones, Faso, et al., 2020; G. L. Williams et al., 2021; but see Heasman & Gillespie, 2018, 2019). Naturally, such interactions do not simulate the more complex interactions between individuals who are known to each other. These interactions may engage higher level social cognitive processes and require greater "emotional engagement" between interactors (Schilbach et al., 2013). Conversely, a greater degree of familiarity between interactors might make social interactions more predictable, thereby lowering the cognitive demands of the interaction. Such contextual factors are likely to interact with each other in complex ways and influence how the DEP operates, just as they do for social cognition more generally. For example, empathic concern—at both cognitive and neural levels of explanation—is known to be contingent upon the strength of the relationship between two individuals (e.g., Hein et al., 2010; Meyer et al., 2013). Moving forward, understanding the boundaries of the DEP theory will require a more precise exploration of how contextual factors, perhaps modeled as statistical moderators, contribute to the DEP (if at all). Overall, we suggest that the information given to, and the relationship between, interaction partners, as well as the topic of conversation underlying the interaction, will be consequential.

Weak Statistical Predictions

> Theories in the "soft areas" of psychology have a tendency to go through periods of initial enthusiasm leading to large amounts of empirical investigation with ambiguous overall results.
>
> —Meehl (1990b, p. 196)

The ultimate stage in a successful derivation chain requires reliable statistical predictions and deriving evidence from empirical data. A strong derivation chain should not only enable directional hypotheses to be falsified but also make clear predictions about the size of any significant effects (i.e., numerical estimates for effect sizes; Meehl, 1990b). Given the multiple chinks in the DEP theory's derivation chain—that is, an unclear theoretical concept, poor measures, and unexplored boundary conditions (Figure 1B)—it is currently impossible to make sufficiently strong statistical predictions about the direction or size of any effects. This issue is compounded by a dearth of open science practices in autism and clinical psychological science (Hobson et al., 2022; Tackett et al., 2019), with almost no published DEP studies having preregistered their hypotheses (apart from Crompton, Ropar, et al., 2020). Taken together, predictions made by the DEP theory are ill-defined and therefore challenging to falsify or corroborate, reflecting wider issues in psychological science (see Scheel, 2022).

Despite a lack of clear hypotheses, we can tentatively derive two broad claims by DEP proponents from existing research, although they have not always been explicitly articulated. Claim 1: Neurotypical people struggle to interact with or understand autistic people, just as autistic people have trouble interacting with or understanding neurotypical people. Claim 2: individuals within the same neurotype (i.e., autistic–autistic, neurotypical–neurotypical) are better at interacting with or understanding one another, compared to mixed neurotype dyads (i.e., autistic–neurotypical). For these claims, we are constrained to using general, imprecise terms (e.g., "interaction" and "understanding"), reflecting the ways in which the DEP theory is underspecified in the literature (see the Fuzzy Theoretical Concept section). In this section, we highlight that the empirical support for even these very general claims is underwhelming.

At first blush, support for Claim 1 stems from studies that use objective measures of social cognition. Brewer et al. (2016) and Edey et al. (2016), for example, found that neurotypical participants were less accurate at identifying social–emotional information—from emotional faces and animated triangles, respectively—generated by autistic compared to neurotypical people. However, inconsistent with Claim 1, autistic people were not reported to be significantly impaired at recognizing facial expressions or mental states generated by neurotypical people. While autistic people are generally found to have emotion recognition and mentalizing difficulties compared to neurotypical individuals (see Velikonja et al., 2019), this pattern of results, required to support Claim 1, was not found when using the stimulus sets in both these studies. Nevertheless, these studies are being widely touted as evidence for the DEP (see below).

Similar problems arise when inspecting the results in Sheppard et al. (2016), which are cited in support of Claim 1 (e.g., Milton et al., 2022; P. Mitchell et al., 2021), although Sheppard et al. (2016) did not explicitly reference the DEP. The authors reported that "neurotypical" observers were more accurate at perceiving neurotypical compared to autistic behavior in video-based stimuli. However, it is unclear whether any analytical steps were taken to ensure that the participants were neurotypical (e.g., screening and/or controlling for observers' autistic traits in the analyses). Further, despite having videos of autistic and neurotypical actors, the authors did not test any autistic participants to examine if they were impaired at recognizing neurotypical individuals' social behaviors. A final issue is that all actors in the videoed stimuli were male, while observers were both male and female, and participant sex was not accounted for in the analyses. Altogether, given that this study was not explicitly designed to test the DEP and, to our knowledge, has never been directly replicated, it does not provide compelling evidence for Claim 1 of the DEP.

Evidence in support of Claim 2 is also equivocal. Some evidence suggests that same versus mixed neurotype social communication is more effective. Crompton, Ropar, et al. (2020) found that communication across mixed neurotype diffusion chains, compared to autistic–autistic and neurotypical–neurotypical chains, was poorer. Similarly, using data from the Crompton, Ropar, et al., 2020 study, Crompton, Sharp, et al. (2020) found mixed neurotype interactors self-reported lower levels of rapport than same neurotype interactors. In Chen et al.'s (2022b) social network study of interactions in a naturalistic setting, autistic and neurotypical individuals showed greater node degree and strength for the same compared to mixed neurotype connections. In Morrison, DeBrabander, Jones, Faso, et al. (2020), where autistic and neurotypical participants rated one another after interacting in a same or mixed neurotype dyad, autistic participants reported there was more information shared with autistic versus neurotypical partners. Confidence in these results should be judged in the context of the design and measurement faults in these studies (see the Measurement Challenges section), but they nevertheless provide some support for Claim 2 of the DEP.

Crucially, however, a closer inspection of the literature in support of Claim 2 shows that many studies do not find enhanced social understanding between individuals in the same compared to mixed neurotype interactions. Most critically, several results do not find improved understanding or interactions between autistic individuals, as often claimed by DEP proponents. For example, Milton et al. (2023) recently argued that the DEP theory suggests "a greater likelihood of feelings of empathy between autistic people with one another" (p. 83). Yet, in Crompton, Sharp, et al. (2020), neurotypical–neurotypical chains reported significantly better rapport than autistic–autistic chains. In Morrison, DeBrabander, Jones, Faso, et al. (2020), *both* neurotypical and autistic participants rated autistic interaction partners as more awkward, less attractive, and less warm than neurotypical partners. Additionally, the autistic participants' interest to interact again with their autistic versus neurotypical conversation partners did not reach statistical significance. Similarly, in Edey et al. (2016), autistic participants were not significantly better at identifying social–emotional information portrayed by other autistic versus neurotypical participants.

Taken together, there are only weak signs of evidence in support of Claims 1 and 2 of the DEP. Beyond these overt problems with the statistical evidence, we cannot help but wonder if findings that contradict the DEP are being obscured or entirely omitted from empirical work. For example, several recent articles inaccurately cite Edey et al. (2016), a study that was never designed to test the DEP, as support for the DEP theory without painting the full picture of results (e.g., Chapman, 2019; Davis & Crompton, 2021; Milton et al., 2023; P. Mitchell et al., 2021; but see Milton et al., 2018). There is also likely to be an impact of publication bias in this field.

This potential cherry-picking of findings may now be feeding back into theoretical discussions and misleading researchers and nonacademic users about the measurement and applications of the DEP. Overall, at the end of the DEP derivation chain, we argue that there is a growing but equivocal set of empirical predictions and findings.

Conclusions and Future Directions

We have presented numerous problems with the DEP theory due to its weak derivation chain. Given the accelerating pace of theoretical and empirical work on the topic, urgent improvements are required. We hope that the current evaluation provides ideas and the impetus for undertaking this important work. While there are numerous avenues that could be pursued in the future, we conclude with four priorities that focus on strengthening the derivation chain and its translation into applied settings: theoretical clarification, improved measures, better statistical approaches, and caution with translation.

Theoretical Clarification

The fundamental problem with the DEP theory concerns conceptual ambiguity and overstretching. Addressing this problem is the main priority for future research, toward a shared conceptual understanding of the DEP. To this end, there are well-established techniques (e.g., Delphi methods; Barrett & Heale, 2020), and even journal article types (e.g., Cubelli & Della Sala, 2017), for developing consensus on concepts. Inclusive approaches that incorporate a diverse range of individuals, across the widest possible range of expertise, offer the best chance of reformulating a solid foundation for the DEP. This should include different stakeholders (researchers, practitioners, individuals with lived experiences), from different disciplines (psychology, social cognitive neuroscience, sociology), with different areas of interest (autism, empathy, wider social cognition, other clinical conditions).

As part of this process, it will be important to broaden the study of the DEP beyond autism research. Indeed, Milton's (2012) initial article was not constrained to autism, yet almost all DEP research has focused on this condition. There is clear scope for studying other neurodevelopmental (e.g., attention-deficit/hyperactivity disorder), neuropsychological (e.g., frontotemporal dementia), and psychiatric (e.g., schizophrenia) conditions—all associated with atypical social behavior and cognition—in relation to the DEP (see also Schilbach et al., 2013). We also recommend that the DEP theory is refined through synthesis with the "second-person neuroscience" literature (see also Manalili et al., 2023). This work is already provoking a serious rethink about traditional theories in social cognitive neuroscience, which have been informed by noninteractive, single-brain studies, in favor of newer models that focus on the interactive elements of social behavior (e.g., Hadley et al., 2022; A. F. D. C. Hamilton & Holler, 2023). Equally, we see untapped value in forging links between the DEP and long-standing concepts in social psychology (e.g., on intergroup relations and bias; Hewstone et al., 2002). This might help to bridge the gap between the DEP's roots in sociology and its study from a cognitive science perspective. Considering whether a renewed DEP theory aligns with predictions made by existing theories (following Scheel, Tiokhin, et al., 2021)—from social neuroscience and social psychology—will help to strengthen the derivation chain for the DEP theory.

Further evaluation and refinement of the DEP theory must grapple with the appropriateness of the term "empathy." Based on the arguments in the current article, there are strong grounds to discontinue the use of the term, and some proponents of the DEP seem to be aware of this issue with terminology (Nicolaidis et al., 2019). Most definitions of the DEP do not actually involve the most widely understood definitions of empathy, and most existing measures of the DEP do not measure empathy itself. While critics might argue that it is just a case of semantics, language and definitions are critically important for descriptive theories in psychology (Eronen & Bringmann, 2021). Additionally, given ongoing debates about empathy more generally (e.g., Bollen, 2023a; Olderbak & Wilhelm, 2020; Stellar & Duong, 2023), it is important that reformulations of the DEP move toward a better synthesis with both the existing and emerging literature on social cognition. As the concept of empathy is refined further in this literature, this will feed into theoretically clarifying the DEP, but only if a more concerted effort is made to do so. It is encouraging to see recent theoretical work starting to rise to this challenge, by suggesting ways to study empathy that could build links with and clarify the concept of the DEP (Bollen, 2023b). On a final note about conceptual clarity, although we have focused on the DEP in relation to autism and social cognition, where most of the literature is focused, there are completely different definitions of "double empathy" in other disciplines (e.g., education; Hargreaves, 2005). This is currently not a major problem, but if this escalates, this confusion will need to be tracked and resolved through more interdisciplinary research.

Improved Measures

We see two general ways to advance the measurement of the DEP, with the caveat that this may not be fruitful until further theoretical development. First, it seems worthwhile trying to mitigate against the problems in the measures that have been used previously (e.g., the diffusion chain paradigm in Crompton, Ropar, et al., 2020; Crompton, Sharp, et al., 2020). We hope that our suggestions for improving tasks in the Measurement Challenges section are a useful starting point. More generally, following emerging recommendations for improving wider psychological science, refining existing measures will necessitate a more rigorous testing of various forms of validity and reliability, the inclusion of control conditions to account for alternative explanations of findings, and replication in larger, more diverse samples (see Nosek et al., 2022). It might seem redundant to make such a simplistic point, but the worrying state of measures in DEP research indicates that this is worth reiterating explicitly as a future direction.

Indeed, this refinement process may not be straightforward. There is a growing awareness of challenges with even the most basic of experimental tasks to measure individual differences in well-established cognitive processes (e.g., attention control; Burgoyne et al., 2023). This is thought to arise from the "reliability paradox" (see Hedge et al., 2018), among other challenges with reconciling discrepancies between measures across different levels of explanation (see Dang et al., 2020). Refining existing and designing new measures to test the DEP will, therefore, benefit from consideration of these contemporary issues in psychological science to move the field toward psychometrically robust measures. Just as measures of social cognition initially developed for use in autism research have advanced social cognitive science more generally (e.g., Frith–Happé Triangles Test; see Livingston et al., 2021), we propose that there is untapped potential for well-designed DEP measures involving autistic and other neurodivergent people to inform theories and methods that will be useful in many other areas of psychological science.

Second, we propose that more promising advances in measurement will come from adopting methods in the cognitive neuroscience literature. Here, research groups have already spent more than a decade theorizing about and testing various social cognitive and dynamic processes involved in two-way interaction (e.g., joint attention and action, social decision making, theory of mind, turn taking, bodily synchrony, and behavioral alignment), as well as adapting neuroscience methods to measure these processes during two-way interaction (see A. F. D. C. Hamilton & Holler, 2023; Redcay & Schilbach, 2019; Sebanz et al., 2006). One method, for example, involves "hyperscanning," which can measure "neural coupling" between two interacting brains (see A. F. D. C. Hamilton, 2021; Pérez & Davis, 2023). This technique has been used to compare dyads, for example, showing reduced neural coupling during interactions between people with and without psychiatric conditions, when compared to dyads of people without any psychiatric conditions (e.g., Bilek et al., 2017). While there are also interesting and serious methodological and technological challenges with studying two-way social neuroscience (see Hadley et al., 2022, for discussion)—for example, reconciling the traditional single-brain model of social cognition with one that incorporates two or more brains—learning lessons from and following developments in cognitive neuroscience is likely to move us toward a more interdisciplinary understanding of the DEP.

There are many other sophisticated developments in measuring social processes that will be useful for advancing the study of the DEP. These include, but are not limited to, capturing synchrony (Paxton & Dale, 2013), dual eye tracking (Pfeiffer et al., 2013), multiuser virtual reality (Pan & Hamilton, 2018), collective psychophysiology (Bolis et al., 2023), machine learning (e.g., Georgescu et al., 2019), and computational modeling (e.g., Lehmann et al., 2023) to investigate social interactions. It is beyond the scope of this article to unpack, in detail, all the possible uses and limitations of these tools. However, in the hope that this article is read by researchers interested in the DEP who may be unfamiliar with these methods, this may be a useful list. We note, for example, one study that captured nonverbal interpersonal synchrony during interaction within different neurotype dyads (autistic–autistic, neurotypical–neurotypical, autistic–neurotypical; Georgescu et al., 2020). The study showed that, compared to neurotypical-neurotypical dyads, there was reduced synchrony in both autistic–neurotypical and autistic–autistic dyads. This indicates no benefit of the same neurotype interaction to autistic people (i.e., tacit evidence against the DEP theory) and, more generally, shows how such methods may be useful to study the DEP in the future.

By taking a more objective, non-autism-specific, and cognitive neuroscience approach to the DEP, we expect that more meaningful progress will be possible. Critically, robust measurement tools will enable us to update our theoretical concepts iteratively, leading to further refinement of our measures. Further, cognitive experimentation will open avenues to develop, test, and refine causal explanations. This will be critical for understanding the mechanisms that underlie the DEP, as well as understanding more distal links with (a)typical behavior and cognition. For example, several DEP proponents have alluded to the possibility that the DEP directly contributes to mental health problems in autism (e.g., Milton et al., 2023; P. Mitchell et al., 2021). There is currently no causal evidence to support this assertion, but this interesting idea warrants investigation. Similarly, a cognitive neuroscience approach will permit better exploration of the developmental origins of the DEP. Most DEP studies, as reflected in the current article, have been conducted in adults and used cross-sectional designs. However, understanding the developmental trajectory of the DEP will be incredibly exciting and important. We advocate for this work being done in tandem with, rather than separately to, current empathy research in developmental science (see Bulgarelli & Jones, 2023), including developmental approaches to second-person neuroscience (e.g., Swain et al., 2013).

Better Statistical Approaches

We reiterate that future research should focus on mending the earliest breaks in the derivation chain, that is, improving the theoretical concept and measurement. There is more of a "validity crisis" (see Eronen & Bringmann, 2021; Schimmack, 2021) and "generalizability crisis" (see Yarkoni, 2022) than a "replication crisis" in DEP research, as well as in empathy and social cognition research more broadly (e.g., Stellar & Duong, 2023; Warnell & Redcay, 2019). Fancy statistical analyses and replication efforts, alone, will not address these problems. Nevertheless, when theories and methods are refined in time, DEP research will benefit from more robust statistical approaches that are starting to advance psychological science more widely. For example, DEP research may be augmented by exploratory techniques, such as multiverse analyses (e.g., Steegen et al., 2016). The parameters for analyses could be varied to test the boundary conditions of the DEP theory and the general robustness of any novel findings (e.g., via sensitivity analyses). DEP studies have also relied almost exclusively on qualitative analysis and/or frequentist null hypothesis significance testing, which Meehl (1978, 1990b) and others have spoken against to ensure meaningful theoretical development. Therefore, greater use of Bayesian analyses (e.g., as per Crompton, Ropar, et al., 2020, in some analyses) or equivalence testing will also help to establish the extent to which there is evidence in favor or against null hypotheses (see Nature Human Behaviour, 2023, for a recent primer).

These suggestions will sound simplistic to the statistically astute reader, but they are worth mentioning given that several findings being cited in relation to the DEP have reported p values around the conventional $p = .05$ significance threshold (e.g., Brewer et al., 2016; Edey et al., 2016; Morrison, DeBrabander, Jones, Faso, et al., 2020). Further, of all the studies discussed in the current article, only Crompton, Ropar, et al. (2020) preregistered their analyses. Therefore, a greater adoption of open science practices, such as preregistration of data collection, data processing pipelines, analyses, and hypotheses, as well as the publication of data and code, will improve the transparency of DEP research. This will be especially important to detect whether there is publication bias (see Scheel, Schijen, & Lakens, 2021). We speculate that null findings opposing the DEP are potentially being "file-drawered" or omitted from the discussion. Once valid and reliable measures are established in the field, meta- and mega-analytic approaches will be useful to quantify and tackle this problem. The use of both Registered and Exploratory

Reports (e.g., Chambers, 2013; McIntosh, 2017) will facilitate progress toward more empirically robust DEP research.

Caution With Translation

In light of the fundamental concerns raised in this article, we conclude by cautioning against the growing number of applications that the DEP is generating in wider society. We are worried that the DEP is widely and uncritically being accepted as a robust explanation of autistic behavior, leading to its application in a variety of settings (e.g., clinical practice, education, policymaking). Suggested changes to clinical practice and interventions are particularly alarming, where there are already discussions of double and even "triple empathy" problems among professionals (Shaw et al., 2023). Given the precarious theoretical and evidence base for the DEP, we argue that carelessly translating the DEP theory into real-world applications may have unintended and potentially harmful consequences for autistic people and those with similar conditions. It might promote intergroup conflict where there is none or lead to distress for some people if within-group interactions are not as successful as the DEP theory predicts they should be. There is a nascent body of work that is starting to speak against the premature translation of "strength-based" interventions for neurodevelopmental conditions without sufficient evidence (e.g., Taylor et al., 2023). Similarly, we conclude this article with a call for caution in any such translation of DEP research until the derivation chain pertaining to this phenomenon is strengthened and the need for such applications is empirically justified.

References

Alkhaldi, R. S., Sheppard, E., & Mitchell, P. (2019). Is there a link between autistic people being perceived unfavorably and having a mind that is difficult to read? *Journal of Autism and Developmental Disorders*, *49*(10), 3973–3982. https://doi.org/10.1007/s10803-019-04101-1

Alkire, D., McNaughton, K. A., Yarger, H. A., Shariq, D., & Redcay, E. (2023). Theory of mind in naturalistic conversations between autistic and typically developing children and adolescents. *Autism: An International Journal of Research and Practise*, *27*(2), 472–488. https://doi.org/10.1177/13623613221103699

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). American Psychiatric Publishing.

Andersen, N. K., Rimvall, M. K., Jeppesen, P., Bentz, M., Jepsen, J. R. M., Clemmensen, L., Jacobsen, R. K., & Olsen, E. M. (2022). A psychometric investigation of the multiple-choice version of Animated Triangles Task to measure Theory of Mind in adolescence. *PLOS ONE*, *17*(3), Article e0264319. https://doi.org/10.1371/journal.pone.0264319

Asghari, S. Z., Farashi, S., Bashirian, S., & Jenabi, E. (2021). Distinctive prosodic features of people with autism spectrum disorder: A systematic review and meta-analysis study. *Scientific Reports*, *11*(1), Article 23093. https://doi.org/10.1038/s41598-021-02487-6

Baron-Cohen, S., Tager-Flusberg, H., & Cohen, D. (2000). *Understanding other minds: Perspectives from autism and developmental cognitive neuroscience*. Oxford University Press.

Barrett, D., & Heale, R. (2020). What are Delphi studies? *Evidence-Based Nursing*, *23*(3), 68–69. https://doi.org/10.1136/ebnurs-2020-103303

Bartlett, F. (1932). *Remembering: A study in experimental and social psychology*. Cambridge University Press.

Bilek, E., Stößel, G., Schäfer, A., Clement, L., Ruf, M., Robnik, L., Neukel, C., Tost, H., Kirsch, P., & Meyer-Lindenberg, A. (2017). State-dependent cross-brain information flow in borderline personality disorder. *JAMA Psychiatry*, *74*(9), 949–957. https://doi.org/10.1001/jamapsychiatry.2017.1682

Bird, G., & Viding, E. (2014). The self to other model of empathy: Providing a new framework for understanding empathy impairments in psychopathy, autism, and alexithymia. *Neuroscience and Biobehavioral Reviews*, *47*, 520–532. https://doi.org/10.1016/j.neubiorev.2014.09.021

Bolis, D., Balsters, J., Wenderoth, N., Becchio, C., & Schilbach, L. (2018). Beyond autism: Introducing the dialectical misattunement hypothesis and a Bayesian account of intersubjectivity. *Psychopathology*, *50*(6), 355–372. https://doi.org/10.1159/000484353

Bolis, D., Dumas, G., & Schilbach, L. (2023). Interpersonal attunement in social interactions: From collective psychophysiology to inter-personalized psychiatry and beyond. *Philosophical Transactions of the Royal Society B*, *378*(1870), Article 20210365. https://doi.org/10.1098/rstb.2021.0365

Bolis, D., Lahnakoski, J. M., Seidel, D., Tamm, J., & Schilbach, L. (2021). Interpersonal similarity of autistic traits predicts friendship quality. *Social Cognitive and Affective Neuroscience*, *16*(1–2), 222–231. https://doi.org/10.1093/scan/nsaa147

Bollen, C. (2023a). A reflective guide on the meaning of empathy in autism research. *Methods in Psychology*, *8*, Article 100109. https://doi.org/10.1016/j.metip.2022.100109

Bollen, C. (2023b). Towards a clear and fair conceptualization of empathy. *Social Epistemology*, *37*(5), 637–655. https://doi.org/10.1080/02691728.2023.2227963

Bora, E., & Pantelis, C. (2016). Meta-analysis of social cognition in attention-deficit/hyperactivity disorder (ADHD): Comparison with healthy controls and autistic spectrum disorder. *Psychological Medicine*, *46*(4), 699–716. https://doi.org/10.1017/S0033291715002573

Bottema-Beutel, K., Kim, S. Y., & Crowley, S. (2019). A systematic review and meta-regression analysis of social functioning correlates in autism and typical development. *Autism Research*, *12*(2), 152–175. https://doi.org/10.1002/aur.2055

Breithaupt, F., Li, B., & Kruschke, J. K. (2022). Serial reproduction of narratives preserves emotional appraisals. *Cognition and Emotion*, *36*(4), 581–601. https://doi.org/10.1080/02699931.2022.2031906

Brewer, R., Biotti, F., Catmur, C., Press, C., Happé, F., Cook, R., & Bird, G. (2016). Can neurotypical individuals read autistic facial expressions? Atypical production of emotional facial expressions in autism spectrum disorders. *Autism Research*, *9*(2), 262–271. https://doi.org/10.1002/aur.1508

Bulgarelli, C., & Jones, E. J. H. (2023). The typical and atypical development of empathy: How big is the gap from lab to field? *JCPP Advances*, *3*(1), Article e12136. https://doi.org/10.1002/jcv2.12136

Burgoyne, A. P., Tsukahara, J. S., Mashburn, C. A., Pak, R., & Engle, R. W. (2023). Nature and measurement of attention control. *Journal of Experimental Psychology: General*, *152*(8), 2369–2402. https://doi.org/10.1037/xge0001408

Burton, A. E., & Harris, R. E. (1947). Case histories in clinical and abnormal psychology. *Harper*.

Camilleri, L. J., Maras, K., & Brosnan, M. (2023). A rule-based theoretical account of social stories to address the double empathy problem. *Frontiers in Psychology*, *14*, Article 1085355. https://doi.org/10.3389/fpsyg.2023.1085355

Camus, L., Macmillan, K., Rajendran, G., & Stewart, M. (2022). *'I too, need to belong': Autistic adults' perspectives on misunderstandings and well-being*. PsyArXiv. https://doi.org/10.31234/osf.io/5mysh

Cañigueral, R., Krishnan-Barman, S., & Hamilton, A. F. C. (2022). Social signalling as a framework for second-person neuroscience. *Psychonomic Bulletin & Review*, *29*(6), 2083–2095. https://doi.org/10.3758/s13423-022-02103-2

Chambers, C. D. (2013). Registered reports: A new publishing initiative at Cortex. *Cortex*, *49*(3), 609–610. https://doi.org/10.1016/j.cortex.2012.12.016

Chapman, R. (2019). Autism as a form of life: Wittgenstein and the psychological coherence of autism. *Metaphilosophy*, *50*(4), 421–440. https://doi.org/10.1111/meta.12366

Chapman, R., & Botha, M. (2023). Neurodivergence-informed therapy. *Developmental Medicine and Child Neurology*, *65*(3), 310–317. https://doi.org/10.1111/dmcn.15384

Chapple, M., Davis, P., Billington, J., Myrick, J. A., Ruddock, C., & Corcoran, R. (2021). Overcoming the double empathy problem within pairs of autistic and non-autistic adults through the contemplation of serious literature. *Frontiers in Psychology*, *12*, Article 708375. https://doi.org/10.3389/fpsyg.2021.708375

Chen, Y. L., Schneider, M., & Patten, K. (2022a). Exploring interpersonal and environmental factors of autistic adolescents' peer engagement in integrated education. *Autism: An International Journal of Research and Practise*, *26*(5), 1255–1266. https://doi.org/10.1177/13623613211046158

Chen, Y. L., Schneider, M., & Patten, K. (2022b). Exploring the role of interpersonal contexts in peer relationships among autistic and non-autistic youth in integrated education. *Frontiers in Psychology*, *13*, Article 946651. https://doi.org/10.3389/fpsyg.2022.946651

Chen, Y. L., Senande, L. L., Thorsen, M., & Patten, K. (2021). Peer preferences and characteristics of same-group and cross-group social interactions among autistic and non-autistic adolescents. *Autism: An International Journal of Research and Practise*, *25*(7), 1885–1900. https://doi.org/10.1177/13623613211005918

Chown, N. (2014). More on the ontological status of autism and double empathy. *Disability & Society*, *29*(10), 1672–1676. https://doi.org/10.1080/09687599.2014.949625

Chown, N., Hughes, L., & Baker-Rogers, J. (2020). What about the other side of double empathy? A response to Alkhaldi, Sheppard and Mitchell's JADD article concerning mind-reading difficulties in autism. *Journal of Autism and Developmental Disorders*, *50*(2), 683–684. https://doi.org/10.1007/s10803-019-04263-y

Christidi, F., Migliaccio, R., Santamaría-García, H., Santangelo, G., & Trojsi, F. (2018). Social cognition dysfunctions in neurodegenerative diseases: Neuroanatomical correlates and clinical implications. *Behavioural Neurology*, *2018*, Article 1849794. https://doi.org/10.1155/2018/1849794

Clutterbuck, R. A., Callan, M. J., Taylor, E. C., Livingston, L. A., & Shah, P. (2021). Development and validation of the Four-Item Mentalising Index. *Psychological Assessment*, *33*(7), 629–636. https://doi.org/10.1037/pas0001004

Clutterbuck, R. A., Livingston, L. A., & Shah, P. (2022). The Four-Item Mentalising Index (FIMI) is a valid, reliable, and practical way to assess mentalising: Reply to Murphy et al. (2022). *Psychological Assessment*, *34*(4), 405–408. https://doi.org/10.1037/pas0001120

Crompton, C. J., DeBrabander, K., Heasman, B., Milton, D., & Sasson, N. J. (2021). Double empathy: Why autistic people are often misunderstood. *Frontiers for Young Minds*, *9*, Article 554875. https://doi.org/10.3389/frym.2021.554875

Crompton, C. J., Hallett, S., Ropar, D., Flynn, E., & Fletcher-Watson, S. (2020). 'I never realised everybody felt as happy as I do when I am around autistic people': A thematic analysis of autistic adults' relationships with autistic and neurotypical friends and family. *Autism: An International Journal of Research and Practise*, *24*(6), 1438–1448. https://doi.org/10.1177/1362361320908976

Crompton, C. J., Ropar, D., Evans-Williams, C. V., Flynn, E. G., & Fletcher-Watson, S. (2020). Autistic peer-to-peer information transfer is highly effective. *Autism: An International Journal of Research and Practise*, *24*(7), 1704–1712. https://doi.org/10.1177/1362361320919286

Crompton, C. J., Sharp, M., Axbey, H., Fletcher-Watson, S., Flynn, E. G., & Ropar, D. (2020). Neurotype-matching, but not being autistic, influences self and observer ratings of interpersonal rapport. *Frontiers in Psychology*, *11*, Article 586171. https://doi.org/10.3389/fpsyg.2020.586171

Cubelli, R., & Della Sala, S. (2017). In search of a shared language in neuropsychology. *Cortex*, *92*, A1–A2. https://doi.org/10.1016/j.cortex.2017.03.011

Dang, J., King, K. M., & Inzlicht, M. (2020). Why are self-report and behavioral measures weakly correlated? *Trends in Cognitive Sciences*, *24*(4), 267–269. https://doi.org/10.1016/j.tics.2020.01.007

Davis, R., & Crompton, C. J. (2021). What do new findings about social interaction in autistic adults mean for neurodevelopmental research? *Perspectives on Psychological Science*, *16*(3), 649–653. https://doi.org/10.1177/1745691620958010

DeBrabander, K. M., Morrison, K. E., Jones, D. R., Faso, D. J., Chmielewski, M., & Sasson, N. J. (2019). Do first impressions of autistic adults differ between autistic and nonautistic observers? *Autism in Adulthood: Challenges and Management*, *1*(4), 250–257. https://doi.org/10.1089/aut.2019.0018

DeNigris, D., Brooks, P. J., Obeid, R., Alarcon, M., Shane-Simpson, C., & Gillespie-Lynch, K. (2018). Bullying and identity development: Insights from autistic and non-autistic college students. *Journal of Autism and Developmental Disorders*, *48*(3), 666–678. https://doi.org/10.1007/s10803-017-3383-y

Dingemanse, M., Liesenfeld, A., Rasenberg, M., Albert, S., Ameka, F. K., Birhane, A., Bolis, D., Cassell, J., Clift, R., Cuffari, E., De Jaegher, H., Novaes, C. D., Enfield, N. J., Fusaroli, R., Gregoromichelaki, E., Hutchins, E., Konvalinka, I., Milton, D., Rączaszek-Leonardi, J., … Wiltschko, M. (2023). Beyond single-mindedness: A figure-ground reversal for the cognitive sciences. *Cognitive Science*, *47*(1), Article e13230. https://doi.org/10.1111/cogs.13230

Doherty, M., McCowan, S., & Shaw, C. K. S. (2023). *Medicine needs all kinds of minds: Autistic doctors are changing culture in healthcare towards a neurodiversity-affirmative approach* [Poster presentation]. It Takes All Kinds Of Minds (ITAKOM) 2023 Conference, Edinburgh, Scotland, United Kingdom.

Edey, R., Cook, J., Brewer, R., Johnson, M. H., Bird, G., & Press, C. (2016). Interaction takes two: Typical adults exhibit mind-blindness towards those with autism spectrum disorder. *Journal of Abnormal Psychology*, *125*(7), 879–885. https://doi.org/10.1037/abn0000199

Eronen, M. I., & Bringmann, L. F. (2021). The theory crisis in psychology: How to move forward. *Perspectives on Psychological Science*, *16*(4), 779–788. https://doi.org/10.1177/1745691620970586

Finke, E. H. (2023). The kind of friend I think I am: Perceptions of autistic and non-autistic young adults. *Journal of Autism and Developmental Disorders*, *53*(8), 3047–3064. https://doi.org/10.1007/s10803-022-05573-4

Fletcher-Watson, S., Brook, K., Hallett, S., Murray, F., & Crompton, C. J. (2021). Inclusive practices for neurodevelopmental research. *Current Developmental Disorders Reports*, *8*(2), 88–97. https://doi.org/10.1007/s40474-021-00227-z

Freese, L., & Burke, P. J. (1994). Persons, identities, and social interaction. M. Barry, H. Karen, & J. O'Brien (Eds.), *Advances in group processes* (Vol. 11, pp. 1–24). JAI. https://wat2140.ucr.edu/PeterBurke/papers/94b.pdf

Frith, U. (2012). Why we need cognitive explanations of autism. *Quarterly Journal of Experimental Psychology: Human Experimental Psychology*, *65*(11), 2073–2092. https://doi.org/10.1080/17470218.2012.697178

Georgescu, A. L., Koehler, J. C., Weiske, J., Vogeley, K., Koutsouleris, N., & Falter-Wagner, C. (2019). Machine learning to study social interaction difficulties in ASD. *Frontiers in Robotics and AI*, *6*, Article 132. https://doi.org/10.3389/frobt.2019.00132

Georgescu, A. L., Koeroglu, S., Hamilton, A. F. C., Vogeley, K., Falter-Wagner, C. M., & Tschacher, W. (2020). Reduced nonverbal interpersonal synchrony in autism spectrum disorder independent of partner diagnosis: A motion energy study. *Molecular Autism*, *11*(1), Article 11. https://doi.org/10.1186/s13229-019-0305-1

Gernsbacher, M. A., & Yergeau, M. (2019). Empirical failures of the claim that autistic people lack a theory of mind. *Archives of Scientific Psychology*, *7*(1), 102–118. https://doi.org/10.1037/arc0000067

Gerring, J. (1999). What makes a concept good? A criterial framework for understanding concept formation in the social sciences. *Polity*, *31*(3), 357–393. https://doi.org/10.2307/3235246

Gonzalez-Liencres, C., Shamay-Tsoory, S. G., & Brüne, M. (2013). Towards a neuroscience of empathy: Ontogeny, phylogeny, brain mechanisms, context and psychopathology. *Neuroscience and Biobehavioral Reviews*, *37*(8), 1537–1548. https://doi.org/10.1016/j.neubiorev.2013.05.001

Green, M. F., Horan, W. P., & Lee, J. (2015). Social cognition in schizophrenia. *Nature Reviews Neuroscience*, *16*(10), 620–631. https://doi.org/10.1038/nrn4005

Grossman, R. B., Mertens, J., & Zane, E. (2019). Perceptions of self and other: Social judgments and gaze patterns to videos of adolescents with and without autism spectrum disorder. *Autism: An International Journal of Research and Practise*, *23*(4), 846–857. https://doi.org/10.1177/1362361318788071

Hadley, L. V., Naylor, G., & Hamilton, A. F. D. C. (2022). A review of theories and methods in the science of face-to-face social interaction. *Nature Reviews Psychology*, *1*(1), 42–54. https://doi.org/10.1038/s44159-021-00008-w

Hamilton, A. F. D. C. (2021). Hyperscanning: Beyond the hype. *Neuron*, *109*(3), 404–407. https://doi.org/10.1016/j.neuron.2020.11.008

Hamilton, A. F. D. C., & Holler, J. (2023). Face2face: Advancing the science of social interaction. *Philosophical Transactions of the Royal Society B*, *378*(1875), Article 20210470. https://doi.org/10.1098/rstb.2021.0470

Hamilton, L. G., & Petty, S. (2023). Compassionate pedagogy for neurodiversity in higher education: A conceptual analysis. *Frontiers in Psychology*, *14*, Article 1093290. https://doi.org/10.3389/fpsyg.2023.1093290

Hand, C. J., Kennedy, A., Filik, R., Pitchford, M., & Robus, C. M. (2023). Emoji identification and emoji effects on sentence emotionality in ASD-diagnosed adults and neurotypical controls. *Journal of Autism and Developmental Disorders*, *53*(6), 2514–2528. https://doi.org/10.1007/s10803-022-05557-4

Happé, F. (2015). Autism as a neurodevelopmental disorder of mind-reading. *Journal of the British Academy*, *3*(1), 197–209. https://doi.org/10.5871/jba/003.197

Happé, F., Cook, J. L., & Bird, G. (2017). The structure of social cognition: In(ter)dependence of sociocognitive processes. *Annual Review of Psychology*, *68*(1), 243–267. https://doi.org/10.1146/annurev-psych-010416-044046

Happé, F., & Frith, U. (2014). Annual research review: Towards a developmental neuroscience of atypical social cognition. *Journal of Child Psychology and Psychiatry, and Allied Disciplines*, *55*(6), 553–577. https://doi.org/10.1111/jcpp.12162

Hargreaves, A. (2005). Educational change takes ages: Life, career and generational factors in teachers' emotional responses to educational change. *Teaching and Teacher Education*, *21*(8), 967–983. https://doi.org/10.1016/j.tate.2005.06.007

Heasman, B., & Gillespie, A. (2018). Perspective-taking is two-sided: Misunderstandings between people with Asperger's syndrome and their family members. *Autism: An International Journal of Research and Practise*, *22*(6), 740–750. https://doi.org/10.1177/1362361317708287

Heasman, B., & Gillespie, A. (2019). Neurodivergent intersubjectivity: Distinctive features of how autistic people create shared understanding. *Autism: An International Journal of Research and Practise*, *23*(4), 910–921. https://doi.org/10.1177/1362361318785172

Hedge, C., Powell, G., & Sumner, P. (2018). The reliability paradox: Why robust cognitive tasks do not produce reliable individual differences. *Behavior Research Methods*, *50*(3), 1166–1186. https://doi.org/10.3758/s13428-017-0935-1

Hein, G., Silani, G., Preuschoff, K., Batson, C. D., & Singer, T. (2010). Neural responses to ingroup and outgroup members' suffering predict individual differences in costly helping. *Neuron*, *68*(1), 149–160. https://doi.org/10.1016/j.neuron.2010.09.003

Henderson, E. N. (1903). A study of memory: For connected trains of thought. *Psychological Monographs*, *5*(6), i–94. https://doi.org/10.1037/h0093003

Hewstone, M., Rubin, M., & Willis, H. (2002). Intergroup bias. *Annual Review of Psychology*, *53*(1), 575–604. https://doi.org/10.1146/annurev.psych.53.100901.135109

Hobson, H., Poole, D., Pearson, A., & Fletcher-Watson, S. (2022). Opening up autism research: Bringing open research methods to our field. *Autism: An International Journal of Research and Practise*, *26*(5), 1011–1013. https://doi.org/10.1177/13623613221105385

House, J. S. (1977). The three faces of social psychology. *Sociometry*, *40*(2), 161–177. https://doi.org/10.2307/3033519

Jellett, R., & Flower, R. L. (2023). How can psychologists meet the needs of autistic adults? *Autism: An International Journal of Research and Practise*. Advance online publication. https://doi.org/10.1177/13623613221147346

Jones, D. R., Morrison, K. E., DeBrabander, K. M., Ackerman, R. A., Pinkham, A. E., & Sasson, N. J. (2021). Greater social interest between autistic and non-autistic conversation partners following autism acceptance training for non-autistic people. *Frontiers in Psychology*, *12*, Article 739147. https://doi.org/10.3389/fpsyg.2021.739147

Kanske, P., & Murray, R. J. (2021). The neurobiological bases of understanding others. *Cortex*, *134*, 351–357. https://doi.org/10.1016/j.cortex.2020.12.012

Kittel, A. F. D., Olderbak, S., & Wilhelm, O. (2022). Sty in the mind's eye: A meta-analytic investigation of the nomological network and internal consistency of the "Reading the Mind in the Eyes" test. *Assessment*, *29*(5), 872–895. https://doi.org/10.1177/1073191121996469

Knoblich, G., & Sebanz, N. (2008). Evolving intentions for social interaction: From entrainment to joint action. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, *363*(1499), 2021–2031. https://doi.org/10.1098/rstb.2008.0006

Kogler, L., Müller, V. I., Werminghausen, E., Eickhoff, S. B., & Derntl, B. (2020). Do I feel or do I know? Neuroimaging meta-analyses on the multiple facets of empathy. *Cortex*, *129*, 341–355. https://doi.org/10.1016/j.cortex.2020.04.031

Lehmann, K., Bolis, D., Friston, K. J., Schilbach, L., Ramstead, M. J. D., & Kanske, P. (2023). An active inference approach to second-person neuroscience. *Perspectives on Psychological Science*. Advance online publication. https://doi.org/10.1177/17456916231188000

Lenzenweger, M. F. (2006). Schizotaxia, schizotypy, and schizophrenia: Paul E. Meehl's blueprint for the experimental psychopathology and genetics of schizophrenia. *Journal of Abnormal Psychology*, *115*(2), 195–200. https://doi.org/10.1037/0021-843X.115.2.195

Lewis, C., & Stack, J. (2013). A mature second-person neuroscience needs a first-person (plural) developmental foundation. *Behavioral and Brain Sciences*, *36*(4), 428–429. https://doi.org/10.1017/S0140525X12001963

Livingston, L. A., Carr, B., & Shah, P. (2019). Recent advances and new directions in measuring theory of mind in autistic adults. *Journal of Autism and Developmental Disorders*, *49*(4), 1738–1744. https://doi.org/10.1007/s10803-018-3823-3

Livingston, L. A., & Happé, F. (2021). Understanding atypical social behaviour using social cognitive theory: Lessons from autism. In H. J. Ferguson & E. E. F. Bradford (Eds.), *The cognitive basis of social interaction across the lifespan* (pp. 147–176). Oxford University Press. https://doi.org/10.1093/oso/9780198843290.003.0007

Livingston, L. A., Shah, P., Milner, V., & Happé, F. (2020). Quantifying compensatory strategies in adults with and without diagnosed autism. *Molecular Autism*, *11*(1), Article 15. https://doi.org/10.1186/s13229-019-0308-y

Livingston, L. A., Shah, P., White, S. J., & Happé, F. (2021). Further developing the Frith-Happé animations: A quicker, more objective, and web-based test of theory of mind for autistic and neurotypical adults. *Autism Research*, *14*(9), 1905–1912. https://doi.org/10.1002/aur.2575

Longo, M. R., & Tsakiris, M. (2013). Merging second-person and first-person neuroscience. *Behavioral and Brain Sciences*, *36*(4), 429–430. https://doi.org/10.1017/S0140525X12001975

Mackes, N. K., Golm, D., O'Daly, O. G., Sarkar, S., Sonuga-Barke, E. J. S., Fairchild, G., & Mehta, M. A. (2018). Tracking emotions in the brain–revisiting the empathic accuracy task. *NeuroImage*, *178*, 677–686. https://doi.org/10.1016/j.neuroimage.2018.05.080

Manalili, M. A. R., Pearson, A., Sulik, J., Creechan, L., Elsherif, M., Murkumbi, I., Azevedo, F., Bonnen, K. L., Kim, J. S., Kording, K., Lee, J. J., Obscura, M., Kapp, S. K., Röer, J. P., & Morstead, T. (2023). From puzzle to progress: How engaging with neurodiversity can improve cognitive science. *Cognitive Science*, *47*(2), Article e13255. https://doi.org/10.1111/cogs.13255

McIntosh, R. D. (2017). Exploratory reports: A new article type for Cortex. *Cortex*, *96*, A1–A4. https://doi.org/10.1016/j.cortex.2017.07.014

Meehl, P. E. (1954). *Clinical versus statistical prediction: A theoretical analysis and a review of the evidence*. University of Minnesota Press. https://doi.org/10.1037/11281-000

Meehl, P. E. (1962). Schizotaxia, schizotypy, schizophrenia. *American Psychologist*, *17*(12), 827–838. https://doi.org/10.1037/h0041029

Meehl, P. E. (1978). Theoretical risks and tabular asterisks: Sir Karl, Sir Ronald, and the slow progress of soft psychology. *Journal of Consulting and Clinical Psychology*, *46*(4), 806–834. https://doi.org/10.1037/0022-006X.46.4.806

Meehl, P. E. (1990a). Toward an integrated theory of schizotaxia, schizotypy, and schizophrenia. *Journal of Personality Disorders*, *4*(1), 1–99. https://doi.org/10.1521/pedi.1990.4.1.1

Meehl, P. E. (1990b). Why summaries of research on psychological theories are often uninterpretable. *Psychological Reports*, *66*(1), 195–244. https://doi.org/10.2466/pr0.1990.66.1.195

Meehl, P. E., & Rosen, A. (1955). Antecedent probability and the efficiency of psychometric signs, patterns, or cutting scores. *Psychological Bulletin*, *52*(3), 194–216. https://doi.org/10.1037/h0048070

Meyer, M. L., Masten, C. L., Ma, Y., Wang, C., Shi, Z., Eisenberger, N. I., & Han, S. (2013). Empathy for the social suffering of friends and strangers recruits distinct patterns of brain activation. *Social Cognitive and Affective Neuroscience*, *8*(4), 446–454. https://doi.org/10.1093/scan/nss019

Milton, D. E. M., Gurbuz, E., & López, B. (2022). The 'double empathy problem': Ten years on. *Autism: An International Journal of Research and Practise*, *26*(8), 1901–1903. https://doi.org/10.1177/13623613221129123

Milton, D. E. M. (2012). On the ontological status of autism: The 'double empathy problem'. *Disability & Society*, *27*(6), 883–887. https://doi.org/10.1080/09687599.2012.710008

Milton, D. E. M. (2018, March 2). *The double empathy problem*. National Autistic Society. https://www.autism.org.uk/advice-and-guidance/professional-practice/double-empathy

Milton, D. E. M., Heasman, B., & Sheppard, E. (2018). Double empathy. In F. Volkmar (Ed.), *Encyclopedia of autism spectrum disorders* (pp. 1–8). Springer. https://doi.org/10.1007/978-1-4614-6435-8_102273-1

Milton, D. E. M., Waldock, K. E., & Keates, N. (2023). Autism and the 'double empathy problem'. In F. Mezzenzana & D. Peluso (Eds.), *Conversations on empathy: Interdisciplinary perspectives on empathy, imagination and othering* (pp. 78–97). Routledge. https://doi.org/10.4324/9781003189978-6

Mitchell, J. P. (2009). Inferences about mental states. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, *364*(1521), 1309–1316. https://doi.org/10.1098/rstb.2008.0318

Mitchell, P., Sheppard, E., & Cassidy, S. (2021). Autism and the double empathy problem: Implications for development and mental health. *British Journal of Developmental Psychology*, *39*(1), 1–18. https://doi.org/10.1111/bjdp.12350

Morrison, K. E., DeBrabander, K. M., Jones, D. R., Ackerman, R. A., & Sasson, N. J. (2020). Social cognition, social skill, and social motivation minimally predict social interaction outcomes for autistic and non-autistic adults. *Frontiers in Psychology*, *11*, Article 591100. https://doi.org/10.3389/fpsyg.2020.591100

Morrison, K. E., DeBrabander, K. M., Jones, D. R., Faso, D. J., Ackerman, R. A., & Sasson, N. J. (2020). Outcomes of real-world social interaction for autistic adults paired with autistic compared to typically developing partners. *Autism: An International Journal of Research and Practise*, *24*(5), 1067–1080. https://doi.org/10.1177/1362361319892701

Murphy, B. A., Hall, J. A., & Duong, F. (2022). It looks like construct validity, but look again: Comment on Clutterbuck et al. (2021) and recommendations for test developers in the broad "empathy" domain. *Psychological Assessment*, *34*(4), 397–404. https://doi.org/10.1037/pas0001063

Nature Human Behaviour. (2023). Points of significance. *Nature Human Behaviour*, *7*, 293–294. https://doi.org/10.1038/s41562-023-01586-w

Nicolaidis, C., Milton, D., Sasson, N. J., Sheppard, E. L., & Yergeau, M. (2019). An expert discussion on autism and empathy. *Autism in Adulthood: Challenges and Management*, *1*(1), 4–11. https://doi.org/10.1089/aut.2018.29000.cjn

Nosek, B. A., Hardwicke, T. E., Moshontz, H., Allard, A., Corker, K. S., Dreber, A., Fidler, F., Hilgard, J., Kline Struhl, M., Nuijten, M. B., Rohrer, J. M., Romero, F., Scheel, A. M., Scherer, L. D., Schönbrodt, F. D., & Vazire, S. (2022). Replicability, robustness, and reproducibility in psychological science. *Annual Review of Psychology*, *73*(1), 719–748. https://doi.org/10.1146/annurev-psych-020821-114157

O'Connor, C., Brassil, M., O'Sullivan, S., Seery, C., & Nearchou, F. (2022). How does diagnostic labelling affect social responses to people with mental illness? A systematic review of experimental studies using vignette-based designs. *Journal of Mental Health*, *31*(1), 115–130. https://doi.org/10.1080/09638237.2021.1922653

Olderbak, S., & Wilhelm, O. (2020). Overarching principles for the organization of socioemotional constructs. *Current Directions in Psychological Science*, *29*(1), 63–70. https://doi.org/10.1177/0963721419884317

Pan, X., & Hamilton, A. F. C. (2018). Why and how to use virtual reality to study human social interaction: The challenges of exploring a new research landscape. *British Journal of Psychology*, *109*(3), 395–417. https://doi.org/10.1111/bjop.12290

Paxton, A., & Dale, R. (2013). Frame-differencing methods for measuring bodily synchrony in conversation. *Behavior Research Methods*, *45*(2), 329–343. https://doi.org/10.3758/s13428-012-0249-2

Pearcey, S., Gordon, K., Chakrabarti, B., Dodd, H., Halldorsson, B., & Creswell, C. (2021). Research Review: The relationship between social anxiety and social cognition in children and adolescents: a systematic review and meta-analysis. *Journal of Child Psychology and Psychiatry, and Allied Disciplines*, *62*(7), 805–821. https://doi.org/10.1111/jcpp.13310

Pérez, A., & Davis, M. H. (2023). Speaking and listening to inter-brain relationships. *Cortex*, *159*, 54–63. https://doi.org/10.1016/j.cortex.2022.12.002

Pfeiffer, U. J., Vogeley, K., & Schilbach, L. (2013). From gaze cueing to dual eye-tracking: Novel approaches to investigate the neural correlates of gaze in social interaction. *Neuroscience and Biobehavioral Reviews*, *37*(10), 2516–2528. https://doi.org/10.1016/j.neubiorev.2013.07.017

Quesque, F., & Rossetti, Y. (2020). What do theory-of-mind tasks actually measure? Theory and practice. *Perspectives on Psychological Science*, *15*(2), 384–396. https://doi.org/10.1177/1745691619896607

Redcay, E., & Schilbach, L. (2019). Using second-person neuroscience to elucidate the mechanisms of social interaction. *Nature Reviews Neuroscience*, *20*(8), 495–505. https://doi.org/10.1038/s41583-019-0179-4

Ricci, L., Lanfranchi, J. B., Lemetayer, F., Rotonda, C., Guillemin, F., Coste, J., & Spitz, E. (2019). Qualitative methods used to generate questionnaire items: A systematic review. *Qualitative Health Research*, *29*(1), 149–156. https://doi.org/10.1177/1049732318783186

Sasson, N. J., Faso, D. J., Nugent, J., Lovell, S., Kennedy, D. P., & Grossman, R. B. (2017). Neurotypical peers are less willing to interact with those with autism based on thin slice judgments. *Scientific Reports*, *7*(1), Article 40700. https://doi.org/10.1038/srep40700

Sasson, N. J., & Morrison, K. E. (2019). First impressions of adults with autism improve with diagnostic disclosure and increased autism knowledge of peers. *Autism: An International Journal of Research and Practise*, *23*(1), 50–59. https://doi.org/10.1177/1362361317729526

Sasson, N. J., Morrison, K. E., Kelsven, S., & Pinkham, A. E. (2020). Social cognition as a predictor of functional and social skills in autistic adults without intellectual disability. *Autism Research*, *13*(2), 259–270. https://doi.org/10.1002/aur.2195

Scheel, A. M. (2022). Why most psychological research findings are not even wrong. *Infant and Child Development*, *31*(1), Article e2295. https://doi.org/10.1002/icd.2295

Scheel, A. M., Schijen, M. R., & Lakens, D. (2021). An excess of positive results: Comparing the standard Psychology literature with Registered Reports. *Advances in Methods and Practices in Psychological Science*, *4*(2), 1–12. https://doi.org/10.1177/25152459211007467

Scheel, A. M., Tiokhin, L., Isager, P. M., & Lakens, D. (2021). Why hypothesis testers should spend less time testing hypotheses. *Perspectives on Psychological Science*, *16*(4), 744–755. https://doi.org/10.1177/1745691620966795

Schilbach, L. (2016). Towards a second-person neuropsychiatry. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *371*(1686), Article 20150081. https://doi.org/10.1098/rstb.2015.0081

Schilbach, L., Timmermans, B., Reddy, V., Costall, A., Bente, G., Schlicht, T., & Vogeley, K. (2013). Toward a second-person neuroscience. *Behavioral and Brain Sciences*, *36*(4), 393–414. https://doi.org/10.1017/S0140525X12000660

Schimmack, U. (2021). The validation crisis in psychology. *Meta-Psychology*, *5*(3), Article MP.2019.1645. https://doi.org/10.15626/MP.2019.1645

Schurz, M., Radua, J., Tholen, M. G., Maliske, L., Margulies, D. S., Mars, R. B., Sallet, J., & Kanske, P. (2021). Toward a hierarchical model of social cognition: A neuroimaging meta-analysis and integrative review of empathy and theory of mind. *Psychological Bulletin*, *147*(3), 293–327. https://doi.org/10.1037/bul0000303

Sebanz, N., Bekkering, H., & Knoblich, G. (2006). Joint action: Bodies and minds moving together. *Trends in Cognitive Sciences*, *10*(2), 70–76. https://doi.org/10.1016/j.tics.2005.12.009

Sebanz, N., & Knoblich, G. (2021). Progress in joint-action research. *Current Directions in Psychological Science*, *30*(2), 138–143. https://doi.org/10.1177/0963721420984425

Shaw, S. C., Carravallah, L., Johnson, M., O'Sullivan, J., Chown, N., Neilson, S., & Doherty, M. (2023). Barriers to healthcare and a 'triple empathy problem'may lead to adverse outcomes for autistic adults: A qualitative study. *Autism*. Advance online publication. https://doi.org/10.1177/13623613231205629

Sheppard, E., Pillai, D., Wong, G. T. L., Ropar, D., & Mitchell, P. (2016). How easy is it to read the minds of people with autism spectrum disorder? *Journal of Autism and Developmental Disorders*, *46*(4), 1247–1254. https://doi.org/10.1007/s10803-015-2662-8

Singer, T., Seymour, B., O'Doherty, J., Kaube, H., Dolan, R. J., & Frith, C. D. (2004). Empathy for pain involves the affective but not sensory components of pain. *Science*, *303*(5661), 1157–1162. https://doi.org/10.1126/science.1093535

Steegen, S., Tuerlinckx, F., Gelman, A., & Vanpaemel, W. (2016). Increasing transparency through a multiverse analysis. *Perspectives on Psychological Science*, *11*(5), 702–712. https://doi.org/10.1177/1745691616658637

Stellar, J. E., & Duong, F. (2023). The little black box: Contextualizing empathy. *Current Directions in Psychological Science*, *32*(2), 111–117. https://doi.org/10.1177/09637214221131275

Strobel, V. (2018, April 14). *Pold87/academic-keyword-occurrence: First release* (Version v1.0.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.1218409

Swain, J. E., Konrath, S., Dayton, C. J., Finegood, E. D., & Ho, S. S. (2013). Toward a neuroscience of interactive parent–infant dyad empathy. *Behavioral and Brain Sciences*, *36*(4), 438–439. https://doi.org/10.1017/S0140525X12002063

Tackett, J. L., Brandes, C. M., King, K. M., & Markon, K. E. (2019). Psychology's replication crisis and clinical psychological science. *Annual Review of Clinical Psychology*, *15*(1), 579–604. https://doi.org/10.1146/annurev-clinpsy-050718-095710

Taylor, E. C., Livingston, L. A., Clutterbuck, R. A., Callan, M. J., & Shah, P. (2023). Psychological strengths and well-being: Strengths use predicts quality of life, well-being and mental health in autism. *Autism: An International Journal of Research and Practise*, *27*(6), 1826–1839. https://doi.org/10.1177/13623613221146440

Velikonja, T., Fett, A. K., & Velthorst, E. (2019). Patterns of nonsocial and social cognitive functioning in adults with Autism Spectrum Disorder: A systematic review and meta-analysis. *JAMA Psychiatry*, *76*(2), 135–151. https://doi.org/10.1001/jamapsychiatry.2018.3645

Warnell, K. R., & Redcay, E. (2019). Minimal coherence among varied theory of mind measures in childhood and adulthood. *Cognition*, *191*, Article 103997. https://doi.org/10.1016/j.cognition.2019.06.009

Weightman, M. J., Air, T. M., & Baune, B. T. (2014). A review of the role of social cognition in major depressive disorder. *Frontiers in Psychiatry*, *5*, Article 179. https://doi.org/10.3389/fpsyt.2014.00179

Williams, G. L., Wharton, T., & Jagoe, C. (2021). Mutual (mis) understanding: Reframing autistic pragmatic "impairments" using relevance theory. *Frontiers in Psychology*, *12*, Article 616664. https://doi.org/10.3389/fpsyg.2021.616664

Williams, Z. J., & Gotham, K. O. (2021). Improving the measurement of alexithymia in autistic adults: A psychometric investigation of the 20-item Toronto Alexithymia Scale and generation of a general alexithymia factor score using item response theory. *Molecular Autism*, *12*(1), Article 56. https://doi.org/10.1186/s13229-021-00463-5

Wu, W., & Mitchell, P. (2019). Accurate inferences of others' thoughts depend on where they stand on the empathic trait continuum. *Personality and Individual Differences*, *148*, 110–116. https://doi.org/10.1016/j.paid.2019.05.025

Yang, C. C., Khalifa, N., & Völlm, B. (2018). The effects of repetitive transcranial magnetic stimulation on empathy: A systematic review and meta-analysis. *Psychological Medicine*, *48*(5), 737–750. https://doi.org/10.1017/S003329171700232X

Yarkoni, T. (2022). The generalizability crisis. *Behavioral and Brain Sciences*, *45*, Article e1. https://doi.org/10.1017/S0140525X20001685

Zaki, J., Bolger, N., & Ochsner, K. (2009). Unpacking the informational bases of empathic accuracy. *Emotion*, *9*(4), 478–487. https://doi.org/10.1037/a0016551

Received April 26, 2023
Revision received October 13, 2023
Accepted December 18, 2023 ■